

NetGuard: A Real-Time Network Attack Detection & Isolation Simulator

INFO 6205 – Program Structure & Algorithms

1. Project Description

NetGuard is a Java Swing based simulation that shows:

- Packet movement through a graph
- Malicious packet movement behaviour
- Real time node infection & isolation
- Dijkstra routing using custom MinHeap
- Automatic graph generation and visualization

The system uses custom ADTs (Bag, MinHeap, LinkedQueue, Graph) to simulate network behavior.

2. Prerequisites

You can run this project using either NetBeans (recommended for GUI) or Eclipse.

A. Java Prerequisite

Install Java JDK 19 or newer

B. NetBeans IDE 22 (Recommended for GUI projects)

C. Eclipse IDE

Notes for Eclipse:

- Eclipse can run the project without issues
- But it cannot edit .form files (GUI layout)
- Editing GUI should be done in NetBeans only

3. Running the Project in NetBeans (Recommended)

1. Open NetBeans
2. Go to File -> Open Project
Select the folder containing:
src/netguard/
nbproject/
build.xml
3. Click Open Project
4. If NetBeans shows “Broken Platform Reference”
 - Right-click Project, select Properties
 - Go to Libraries , select Java Platform
 - Select JDK 19 or higher
5. Run the project:
 - Right-click Netguard.java → Run File
OR
 - Press the green -> Run Project button
 - The NetGuard GUI window opens.

4. Running the Project in Eclipse

1. Open Eclipse
2. Go to File -> Import
3. Select General -> Existing Projects into Workspace
4. Browse to the NetGuard folder
5. Click Finish

Run the project:

Find file: src/netguard/Netguard.java

Right click -> Run As -> Java Application

Important Eclipse notes:

- Eclipse cannot interpret .form GUI files
- GUI design may not show preview
- But the program runs perfectly
- All logic (GraphPanel, Graph, MinHeap, Packet) works identical

5. How to Use the Application

Start

- Begins simulation
- Generates 3 random packets
- 25% chance each is malicious
- Routes packets using Dijkstra

Pause / Resume

- Stops animation timer
- Maintains current state
- Resumes from exactly where it paused

Graph

- Generates a new random graph (10–15 nodes)
- Resets packets, infections, isolated nodes

Status Panel

Shows:

- Active Packets
- Infected Nodes
- Isolated Nodes
- Simulation State

Legend Panel

Explains:

- Healthy nodes (green)
- Infected nodes (orange)
- Isolated nodes (gray)
- Safe packets (blue)
- Malicious packets (red)

6. Known Limitations

- Large graphs (>20 nodes) reduce FPS
- Eclipse cannot edit .form files
- Infection cascade timers can overlap if many packets hit at once
- GraphPanel resolution optimized for 900×700 window